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Name _____

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Department _____

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University Physics

Final Examination

School of Intensive Instruction in Sciences and Arts, Nanjing University

Select five out of following six problems.

1. (20 pts.) A stick of proper length l sits at rest in the frame S, lying in the xy plane at an angle of $\theta = \arctan(3/4)$ with the x -axis. A frame S' moves with velocity $\mathbf{v} = v\mathbf{e}_x$ with respect to S. In the frame S', the stick is angled at 45° with respect to the x' -axis.

(a) What is v ?

(b) What is the length l' of the rod as measured in the frame S'?

Solution: (a) Assume that in the frame S, at $t = 0$, one end of the stick is at the origin and the other end is at point (x, y) , and in the frame S', at $t' = 0$, one end of the stick is at the origin and the other end is at point (x', y') . Then, we have

$$y' = y, \quad x' = \frac{x}{\gamma}.$$

Therefore,

$$\tan \theta' = \frac{y'}{x'} = \gamma \frac{y}{x} = \gamma \tan \theta, \quad \gamma = \frac{\tan \theta'}{\tan \theta} = \frac{4}{3}, \quad \beta = \sqrt{1 - \frac{1}{\gamma^2}} = \frac{\sqrt{7}}{4}, \quad v = \frac{\sqrt{7}}{4}c.$$

$$(b) \quad l' = \sqrt{\frac{x'^2}{\gamma^2} + y'^2} = \sqrt{x^2 + y^2 - \beta^2 x^2} = \sqrt{l^2 - l^2 \beta^2 \cos^2 \theta} = l \sqrt{1 - \beta^2 \cos^2 \theta}.$$

$$\cos^2 \theta = \frac{1}{1 + \tan^2 \theta} = \frac{16}{25}.$$

$$l' = l \sqrt{1 - \frac{7}{25}} = \frac{3\sqrt{2}}{5}l.$$

2. (20 pts.) The internal energy of systems 1 and 2 is given by $U_i = C_i T$ ($i=1,2$), where C_i is the constant heat capacity of system i . Originally, system 1 is at temperature T_1 and system 2 is at temperature T_2 . The two systems are brought in thermal contact with each other but are completely isolated to the rest of the world. The total system will finally reach equilibrium at common temperature T_f .

(a) Calculate T_f .

(b) Calculate the change in entropy during this process.

(c) Show that the entropy has increased.

Solution: (a)

$$C_1 T_1 + C_2 T_2 = (C_1 + C_2) T_f, \quad T_f = \frac{C_1 T_1 + C_2 T_2}{C_1 + C_2}.$$

(b)

$$\Delta S_t = \int_{T_1}^{T_f} \frac{C_1 dT}{T} + \int_{T_2}^{T_f} \frac{C_2 dT}{T} = (C_1 + C_2) \ln T_f - C_1 \ln T_1 - C_2 \ln T_2.$$

(c) Since the logarithm is the concave function, then we have

$$\ln(cx + (1-c)y) > c \ln x + (1-c) \ln y \quad \text{for } 0 < c < 1.$$

Thus,

$$\begin{aligned}\Delta S_i &= (C_1 + C_2) \ln T_f - C_1 \ln T_1 - C_2 \ln T_2 \\ &= (C_1 + C_2) \left[\ln \left(\frac{C_1}{C_1 + C_2} T_1 + \frac{C_2}{C_1 + C_2} T_2 \right) - \frac{C_1}{C_1 + C_2} \ln T_1 - \frac{C_2}{C_1 + C_2} \ln T_2 \right] > 0.\end{aligned}$$

3. (20 pts.) An ideal gas initially at a pressure p_0 undergoes a free expansion until its final volume is 8 times its initial volume V_0 . The gas is then slowly and adiabatically compressed back to its initial volume at which the pressure is $2^{6/5} p_0$.

(a) Determine whether the ideal gas is monatomic, diatomic or polyatomic.

(b) Find the total change in entropy in this process.

Solution: (a) In the adiabatic process, we have

$$\frac{p_0}{8} (8V_0)^\gamma = 2^{6/5} p_0 (V_0)^\gamma.$$

Then

$$8^{\gamma-1} = 2^{6/5}, \quad \gamma - 1 = \frac{2}{5}, \quad \gamma = \frac{7}{5}. \quad (C_V = \frac{nR}{\gamma-1} = \frac{5}{2} nR).$$

Therefore, the gas is diatomic.

(b) The change in entropy in the adiabatic process is zero. Then imaging a reversible isothermal process connecting the states p_0, V_0 and $p_0/8, 8V_0$, we obtain

$$\Delta S = \int_{V_0}^{8V_0} \frac{pdV}{T} = \int_{V_0}^{8V_0} \frac{nRdV}{V} = nR \ln 8.$$

4. (20 pts.) The Van der Waals gas of one mole has the equation of the state

$$\left(p + \frac{a}{V^2} \right) (V - b) = RT.$$

(a) Discuss the physical meaning of the parameters a and b . Why is there the correction to the pressure proportional to $1/V^2$?

(b) The gas undergoes an isothermal expansion from the volume V_1 to the volume V_2 . Calculate the change in the Helmholtz free energy.

(c) Find the change in the internal energy for the expansion of (b).

Solution: (a) b is the closed packed volume of one mole of the molecules.

a is due to the attractive interaction between the molecules.

The correction is due to the attractive interaction between the molecules.

(b) Since

$$dF = -SdT - pdV.$$

In an isothermal expansion, the change in Helmholtz free energy is

$$\Delta F = - \int_{V_1}^{V_2} pdV = - \int_{V_1}^{V_2} \left(\frac{RT}{V-b} - \frac{a}{V^2} \right) dV = -RT \ln \frac{V_2-b}{V_1-b} + a \left(\frac{1}{V_1} - \frac{1}{V_2} \right).$$

(c) From

$$dU = TdS - pdV = T \left(\frac{\partial S}{\partial T} \right)_V dT + \left[T \left(\frac{\partial S}{\partial V} \right)_T - p \right] dV,$$

and using the Maxwell relation

$$\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial p}{\partial T} \right)_V$$

we have

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p = \frac{a}{V^2}.$$

Therefore, in the isothermal expansion

$$\Delta U = a\left(\frac{1}{V_1} - \frac{1}{V_2}\right).$$

5. (20 pts.) For a classical ideal gas the number of molecules having the components of velocities between v_x and $v_x + dv_x$, v_y and $v_y + dv_y$ and v_z and $v_z + dv_z$ is

$$f(v_x, v_y, v_z) dv_x dv_y dv_z = N \left(\frac{m}{2\pi k_B T} \right)^{3/2} \exp\left(-\frac{1}{2} m(v_x^2 + v_y^2 + v_z^2) / k_B T\right) dv_x dv_y dv_z,$$

where N is the total number of molecules.

(a) Find the number of molecules having speeds between v and $v + dv$.

(b) Calculate the average kinetic energy of the molecules.

$$(I_{2m} = \int_0^\infty x^{2m} e^{-x^2} dx = \frac{1}{2} \frac{1}{2} \frac{3}{2} \dots \frac{2m-1}{2} \sqrt{\pi}.)$$

Solution: (a) The number of molecules having speeds between v and $v + dv$ is

$$f(v)dv = N \left(\frac{m}{2\pi k_B T} \right)^{3/2} \exp\left(-\frac{1}{2} m v^2 / k_B T\right) 4\pi v^2 dv.$$

(b) The mean time between collisions for molecules of all speeds is

$$\begin{aligned} \bar{T} &= \frac{\int_0^\infty \frac{1}{2} m v^2 f(v) dv}{N} = \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty \frac{1}{2} m v^2 \exp\left(-\frac{1}{2} m v^2 / k_B T\right) 4\pi v^2 dv \\ &= \left(\frac{m}{2\pi k_B T} \right)^{3/2} 2\pi m \left(\frac{2k_B T}{m} \right)^{5/2} \int_0^\infty x^4 e^{-x^2} dx \\ &= \frac{4k_B T}{\sqrt{\pi}} \cdot I_4 = \frac{4k_B T}{\sqrt{\pi}} \frac{3}{8} \sqrt{\pi} = \frac{3}{2} k_B T. \end{aligned}$$

6. (20 pts.) In the neighborhood of the triple point of ammonia (NH_3), the equation for the sublimation curve is $\ln p = 27.79 - 3726/T$ and the equation for the vaporization curve is $\ln p = 24.10 - 3005/T$ where p is measured in pascals and T is measured in kelvins.

(a) Compute the temperature and pressure of the triple point.

(b) What is the molar latent heat of sublimation? What is the molar latent heat of vaporization?

(The molar gas constant is $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$.)

Solution: (a) $\ln p = 27.79 - 3726/T$ (1)

$\ln p = 24.10 - 3005/T$ (2)

(1)-(2)

$$0 = 3.69 - \frac{721}{T}, \quad T = 195.4 \text{ K}.$$

Substituting $T = 195.4 \text{ K}$ into Eq. 1, we have

$$\ln p = 27.79 - 3726/T = 8.72.$$

Therefore,

$$p = 6.12 \times 10^3 \text{ Pa}$$

(2) Differentiating both sides of Eqs.1 and 2, we obtain

$$\frac{dp}{p} = \frac{3726}{T^2} dT ,$$

$$\frac{dp}{p} = \frac{3005}{T^2} dT .$$

Noting that $pV = RT$, we have

$$\frac{dp}{dT} = \frac{3726R}{TV} ,$$

$$\frac{dp}{dT} = \frac{3005R}{TV} .$$

Thus, the molar latent heat of sublimation is $L_s = 3726R = 3.1 \times 10^4 \text{ J} \cdot \text{mol}^{-1}$. The molar latent heat of vaporization is $L_v = 3005R = 2.5 \times 10^4 \text{ J} \cdot \text{mol}^{-1}$.